Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years developing reliable **C++**, **C#**, and Python systems through object-oriented design, automated **verification and validation**, CI/CD, cross-platform testing, debugging, documentation, and quality-focused delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Software Engineering, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Structures, Computer Networks, Web Technologies

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built fault-tolerant application workflows with explicit job states, validation, and failure isolation, processing 1K+ concurrent jobs at triple throughput while sustaining 99%+ **software reliability** across 5K+ monthly production tasks.
- Developed normalized data services with access controls, reconciliation checks, and indexed queries for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through **performance debugging and optimization**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Developed application interfaces and backend services for secure uploads, grading history, and feedback, supporting 150+ instructors and 10K+ daily requests while applying **system integration testing** across production workflows.
- Implemented concurrent batch processing with error isolation, provider fallbacks, and output validation, executing 1K+ files per run while automated release checks reduced feedback turnaround 60% and strengthened **defect prevention**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*
- Built reproducible Python test pipelines processing 100K+ autonomous-driving trajectory logs, improving planning 18% while conducting **software verification and validation** through controlled simulations, baseline comparisons, and documented results.
- Evaluated a spatial-temporal encoder through automated closed-loop system tests, improving stability 15% while investigating failures, measuring performance tradeoffs, and formally communicating **system-level test evidence** across research reviews.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Engineered cross-platform software with object-oriented Singleton and Factory patterns, improving API response times 30%, reducing technical debt 20%, and strengthening **maintainability** across Windows, Linux, and macOS deployments.
- Automated 20+ **unit and integration tests** with PyTest, Mockito, Jenkins, and virtual-machine builds across three operating systems, supporting defect resolution, Agile reviews, and release validation with external stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Built eight **C# and .NET** modules using object-oriented design for authentication, resources, billing, leads, permissions, and notifications, reliably synchronizing 2M+ records across distributed locations with real-time updates.
- Implemented automated unit, integration, and end-to-end tests through Jenkins CI/CD and Docker staging, safely reaching 95%+ **build validation** and reducing deployment time 50% across controlled release cycles.

Skills

Languages: C++, C#, Python, Java, SQL, TypeScript, JavaScript
Systems: Qt, .NET, GUI Development, Embedded Linux, Linux, Windows, OOP, SOLID, Design Patterns, CMake, Git, System Integration
Testing: Unit Testing, Integration Testing, System Testing, Verification and Validation, Test Procedures, Regression Testing, PyTest, Debugging
Delivery & Quality: Jenkins, CI/CD, Docker, Build Automation, Version Control, Technical Documentation, Code Review, Requirements Analysis, Risk Analysis, Secure Development, Cybersecurity, Agile/Scrum, Quality Compliance

Projects

Device Verification Console | *C++, Qt, Embedded Linux, CMake, GoogleTest, Python*
- Designed a Qt-based **medical-device verification console** that exercises simulated device commands, records traceable results, detects out-of-range responses, and generates repeatable unit, integration, and system-level test reports for regulated workflows.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*
- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **scheduling algorithms** in C++ with interactive visualization, validating 1K+ process combinations and reducing average turnaround time up to 25%.

HMIS – Hospital Management Information System Live | *MERN, TypeScript, ABDM APIs, Sentry*
- Built secure healthcare **workflow software** with consent-based patient-history access, appointment management, Twilio notifications, and Sentry monitoring, reducing administrative workload 30% while supporting 100+ daily appointments through reliable monitoring.